

EXPERIMENT NO. 12:

Implement Iris Flower Classification using KNN

Aim:

To implement Iris Flower Classification using the K-Nearest Neighbours (KNN) algorithm in Python.

Procedure:

1. Import the required Python libraries.
2. Load the Iris dataset.
3. Separate the input features and target classes.
4. Split the dataset into training and testing sets.
5. Create a KNN classifier with a suitable value of K.
6. Train the model using the training dataset.
7. Predict the flower species for the test data.
8. Calculate and display the classification accuracy.
9. Predict the species of a new Iris flower sample.

Program:

```
# Iris Flower Classification using KNN

from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.neighbors import KNeighborsClassifier

from sklearn.metrics import accuracy_score

# Load Iris Dataset

iris = load_iris()

X = iris.data

y = iris.target

# Split dataset into training and testing sets

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.3, random_state=42

)

# Create KNN Classifier

knn = KNeighborsClassifier(n_neighbors=3)

# Train the model
```

```
knn.fit(X_train, y_train)

# Predict test data

y_pred = knn.predict(X_test)

# Calculate accuracy

accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:")

print(round(accuracy * 100, 2), "%")

# Predict a new Iris flower

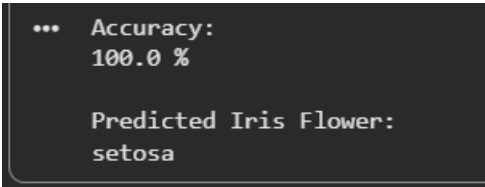
new_flower = [[5.1, 3.5, 1.4, 0.2]]

prediction = knn.predict(new_flower)

print("\nPredicted Iris Flower:")

print(iris.target_names[prediction][0])
```

OUTPUT:

A terminal window with a dark background and light green text. It shows the output of the Python code: 'Accuracy: 100.0 %' followed by 'Predicted Iris Flower: setosa' on a new line.

```
... Accuracy:
100.0 %

Predicted Iris Flower:
setosa
```

Result:

The Iris Flower Classification program was successfully implemented using the K-Nearest Neighbours (KNN) algorithm. The model classified the new Iris flower sample as Setosa and achieved high classification accuracy.